Background reading · Plague, marmots, and the year 1910

Life Sciences · ages 15–18 · for the homework question "Why 1910 and not 1890?"

The slides gave you the shape of the story. This gives you the evidence. Everything below is dated and sourced. The argument is yours to build — it is not written down here.

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1 · Plague is not one disease

Yersinia pestis causes three different illnesses, and which one you get depends on how the bacterium enters you. The figures below are from the World Health Organization.

Bubonic. A flea bites you. The bacteria travel to the lymph nodes, which swell into painful lumps called buboes. WHO puts the case fatality ratio at **30–60%**. **Human-to-human transmission of bubonic plague is rare** — the flea is what moves it from one person to the next.

Septicaemic. The bacteria multiply rapidly in the bloodstream.

Pneumonic. The bacteria reach the lungs. WHO calls this the **most virulent form**. The sick person coughs, and respiratory particles carry live bacteria to anyone nearby. **No flea is involved at all.** Untreated, it can be fatal **within 18 to 24 hours** of symptoms starting, and the incubation period can be as short as 24 hours.

That last form is the one that matters here. Once plague is in a person's lungs it has stopped being an animal disease that occasionally spills over into people. It has become a human respiratory epidemic, and it spreads wherever people breathe the same air.

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2 · Where plague lives when nobody is dying of it

Plague does not appear and disappear. It lives permanently in populations of wild rodents. Scientists call such a place a **natural focus**.

The mountain steppe of southern Siberia and Mongolia is one of these. The host there is the **tarbagan marmot** (*Marmota sibirica*), a large burrowing rodent that lives in colonies of connected tunnels.

How long has it been there?

- Researchers screened DNA from the teeth of **101 people who died in the Bronze Age** across Eurasia and found *Y. pestis* in **seven** of them. The two oldest are from burials in **southern Siberia, roughly 2900–2700 BC** (Rasmussen et al., *Cell*, 2015).
- Russian teams still sequence it there today: **33 strains collected in Gorny Altai between 2012 and 2017** (Eroshenko et al., 2018).

So the bacterium was in these mountains before the pyramids were finished, and it is in them now. **Nothing about the reservoir changed between 1890 and 1910.**

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3 · How a person catches it from a marmot

A hunter who skins an infected marmot can be infected through cuts on the hands, or by breathing in bacteria from the animal. Fleas leaving the dead animal can bite.

Local hunters in the region — many of them Buryat — knew this. They hunted healthy, fast animals by stalking them. **A marmot that is slow and easy to catch is a sick marmot**, and the traditional rule was to leave it alone, and to move camp if sick animals appeared. That knowledge is not written in a textbook; it is passed from hunter to hunter.

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4 · Manchuria in 1890 and in 1910

Facts only. What you do with them is the assignment.

	around 1890	by 1910
Railway across Manchuria	None. The Chinese Eastern Railway had not been built.	Opened **1 June 1903**. T-shaped, meeting at Harbin. Manzhouli sits on the western branch. The southern branch passed to Japan in 1905 and became the South Manchuria Railway.
What marmot fur was worth	No figures recorded. The documented price rise begins after 1907.	Rose roughly **six-fold between 1907 and 1910**: about 15 kopecks a skin, to 50 in 1909, to 80 in 1910. One contemporary source gives a wider range, from 20 kopecks to 1.20 roubles.
Why the price moved	—	Dyers in **Leipzig** worked out how to blacken cheap marmot fur so it could be sold as imitation sable, which was fashionable.
Who was hunting	Local hunters, at local scale, using traditional methods.	**Thousands of Chinese migrant hunters arrived after 1909.** The population of Manzhouli reached about **10,000**, mostly seasonal trappers. Most had never hunted a marmot before.
Where the fur went	Local and regional use.	By rail to **Moscow and Leipzig**.
Where the hunters slept	—	Packed together in inns and dugouts through a Manchurian winter.

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5 · What happened, autumn 1910 to spring 1911

- The first case was recorded at **Manzhouli** in autumn 1910.
- By **27 October** the disease had reached **Harbin**, the railway junction.
- It travelled along the rail lines. Chinese New Year was approaching and migrant workers were going home on cheap third-class tickets. Cities on the line were hit first — Harbin, Changchun, Mukden — then smaller places off it. Cases reached **Beijing and Tianjin**.
- The last recorded case was **1 March 1911**. Authorities declared the outbreak over at the end of April.
- **More than 60,000 people died.** The death rate among those infected was close to **100%**.

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6 · What Wu Lien-teh established

Wu Lien-teh was a Cambridge-trained physician sent to Harbin. At the time, most experts assumed this was ordinary bubonic plague carried by rats and fleas.

He carried out post-mortem examinations and showed that the disease was in the **lungs**. It was **pneumonic**, passing directly from person to person in the air.

That finding changed what to do. If fleas are the route, you kill rats. If breath is the route, you separate people and cover faces. He introduced a gauze-and-cotton mask, strict quarantine, and "plague passports" for people released from it.

In **April 1911** he presided over the **International Plague Conference at Mukden** — about 33 delegates from 11 countries, the first international scientific congress held in China.

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7 · Your task

The slide said: *the bacterium did not change, the network did.* Your job is to show **which** parts of the network changed, **when**, and **why that produced an epidemic**.

Work through these as you write:

1. **List what is different between the two columns in section 4.** There is more than one thing.
2. **Test each one.** For each change you list, ask: if this one thing had not happened, would 60,000 people still have died? Some of your changes will turn out to be essential, others will not.
3. **Separate two different questions.** What caused people to *become infected* in the first place? And what caused the disease to *travel hundreds of kilometres*? These may not have the same answer.
4. **Explain the switch in section 1.** Plague in a marmot colony is a flea-borne disease. Sixty thousand people did not die of flea bites. Where in your story does the disease become pneumonic, and what made that possible?
5. **Rule one explanation out before you begin.** The same bacterium lived in the same marmots in 1890 and in 1910. Explain why that fact alone is enough to reject "the bacterium became more dangerous" as the answer — and notice that you do not need a single piece of evidence about 1910 to do it.

Write **400–600 words**. Use dates and numbers from this sheet.

Sources

- World Health Organization, Plague fact sheet — forms of the disease, case fatality, transmission, incubation.
- Rasmussen S, et al. Early divergent strains of *Yersinia pestis* in Eurasia 5,000 years ago. *Cell* 2015;163(3):571–582.
- Eroshenko GA, et al. Natural mega-focus of *Yersinia pestis*, phylogenetic line 4.ANT in Gorny Altai. *Problems of Particularly Dangerous Infections* 2018;(2):49–56.
- Shen. Pneumonic Plagues, Environmental Changes, and the International Fur Trade: The Retreat of Tarbagan Marmots from Northwest Manchuria, 1900s–30s.
- The 1910–1911 Manchurian plague and Wu Lien-teh. *Le Infezioni in Medicina* 2022;30(3):464–468.

- disasterhistory.org, "The Manchurian plague, 1910–11".